

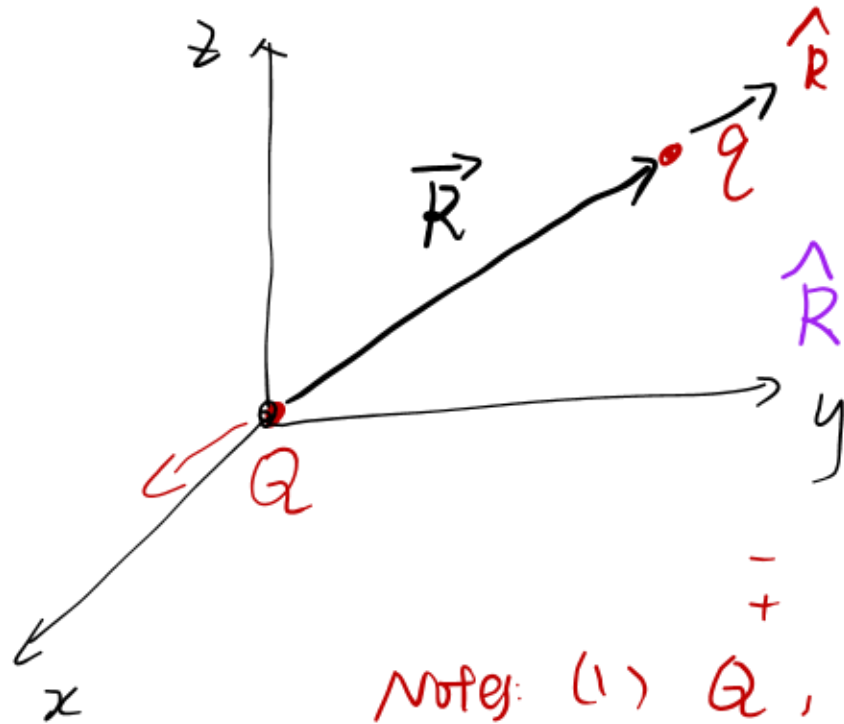


Lecture 5

- Coulomb's law, some examples
- Gauss's law



Coulomb's law describes the amount of force between two electrically charged particles at rest.



$$\hat{R} = \frac{\vec{R}}{|\vec{R}|}$$

$$\vec{F}_{Q \rightarrow q}$$

$$= \frac{1}{4\pi\epsilon} \cdot \frac{Qq}{|\vec{R}|^2} \hat{R}$$

permittivity
in free space $\epsilon = \epsilon_0 = 8.854 \times 10^{-12} \left[\frac{F}{m} \right]$
material $\epsilon = \epsilon_0 \epsilon_r$

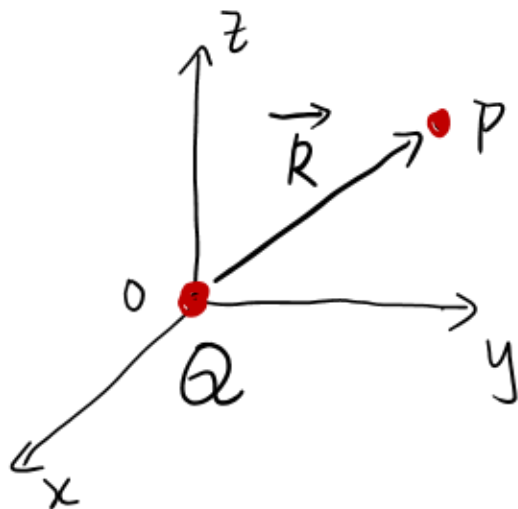
distance between Q and q

Notes: (1) $\begin{matrix} - & - & + \\ + & + & + \end{matrix}$ Q, q Qq force is repulsive
 $\begin{matrix} + & - & - \\ + & - & - \end{matrix}$ Q, q Qq force is attractive

$$(2) \vec{F}_{q \rightarrow Q} = - \vec{F}_{Q \rightarrow q}$$

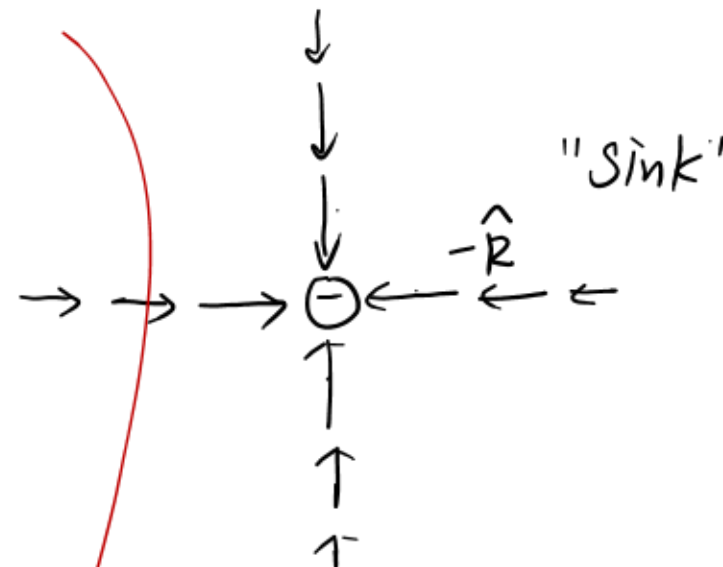
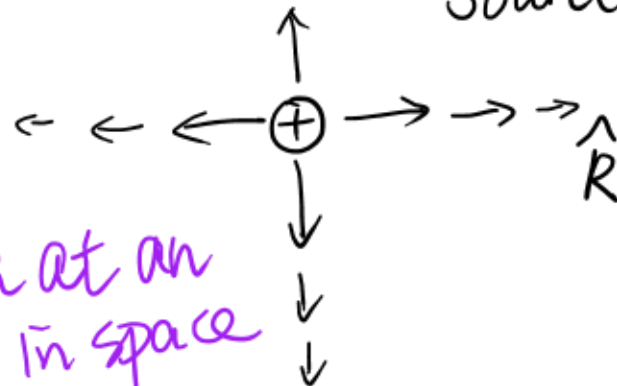
(3) Force decreases with distance $\frac{1}{|\vec{R}|^2}$

Coulomb's law for electric field

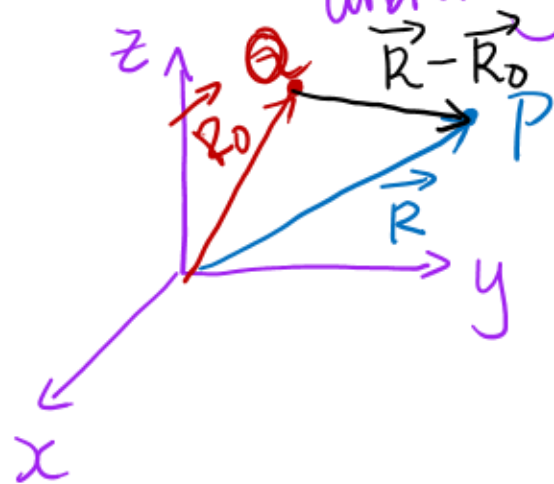


$$\vec{E} = \frac{\vec{F}}{q} = \frac{1}{4\pi\epsilon} \frac{Q}{|\vec{R}|^3} \vec{R} \left[\frac{N}{C} \right]$$

"source"



Generalize to Q at an arbitrary point in space



(point of observation)

$\vec{E}$ produced by Q at P =

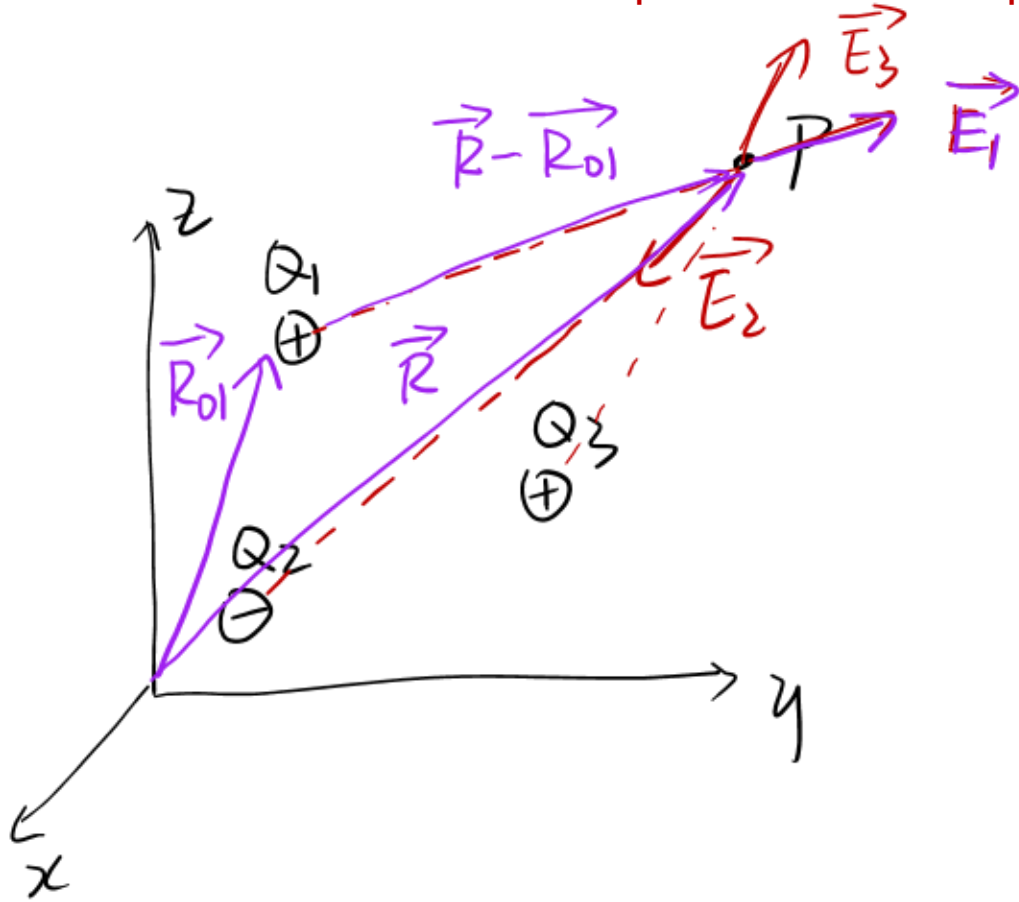
$$\vec{E} = \frac{1}{4\pi\epsilon} \frac{Q}{|\vec{R} - \vec{R}_0|^3} (\vec{R} - \vec{R}_0) \left[\frac{N}{C} \right]$$

source

point of observation

source

Calculate the electric field at point P from multiple point charges



"Superposition"

$$\vec{E} = \frac{1}{4\pi\epsilon} \frac{Q}{|\vec{R} - \vec{R}_0|^3} (\vec{R} - \vec{R}_0)$$

$$\vec{E}_1 = \frac{1}{4\pi\epsilon} \frac{Q_1}{|\vec{R} - \vec{R}_{01}|^3} (\vec{R} - \vec{R}_{01})$$

$$\vec{E}_2 = \frac{1}{4\pi\epsilon} \frac{Q_2}{|\vec{R} - \vec{R}_{02}|^3} (\vec{R} - \vec{R}_{02})$$

$$\vec{E}_3 = \frac{1}{4\pi\epsilon} \frac{Q_3}{|\vec{R} - \vec{R}_{03}|^3} (\vec{R} - \vec{R}_{03})$$

$$\vec{E}_{\text{total}} = \vec{E}_1 + \vec{E}_2 + \vec{E}_3$$

Calculate the electric field at point P from multiple point charges



Steps of using Coulomb's law to calculate $\vec{E}$ field

1. Choose coordinate system. define origin
2. Write radius vector of the Source (charge) $\vec{R}_0$
3. Write radius vector of point of observation $P = \vec{R}$
4. Find vector from Source to point $\vec{R} - \vec{R}_0$
5. Apply Coulomb's law
6. Sum or Integrate to find $\vec{E}$ at point of observation P.
 $\Downarrow$ point source $\Downarrow$ charge distribution

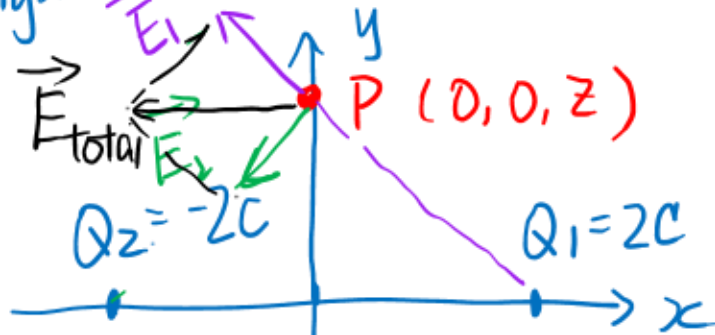


Calculate the electric field at point P from multiple point charges [Example]

Calculate the electric field at point P on the z-axis from two point charges:

$Q_1 = 2C$, $Q_2 = -2C$
located at
 $(3, 0, 0)m$ and $(-3, 0, 0)m$,
x y z respectively.

Physical understanding =



E_{total} should be in $-\hat{x}$ direction.

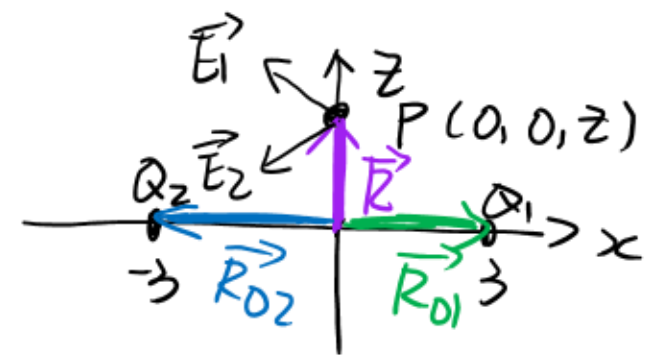
agree

Calculation:

For Q_1 = $\vec{R} = \hat{z}z$
 $\vec{R}_{01} = \hat{x}3$
 $\vec{R} - \vec{R}_{01} = \hat{z}z - \hat{x}3$
 $|\vec{R} - \vec{R}_{01}| = \sqrt{z^2 + 9}$

$$\vec{E}_1 = \frac{1}{4\pi\epsilon_0} \frac{Q_1}{|\vec{R} - \vec{R}_{01}|^3} (\vec{R} - \vec{R}_{01})$$

$$= \frac{1}{4\pi\epsilon_0} \frac{2}{(z^2 + 9)^{\frac{3}{2}}} (\hat{z}z - \hat{x}3)$$



For Q_2 = $\vec{R} = \hat{z}z$
 $\vec{R}_{02} = -\hat{x}3$
 $\vec{R} - \vec{R}_{02} = \hat{z}z + \hat{x}3$
 $|\vec{R} - \vec{R}_{02}| = \sqrt{z^2 + 9}$

$$\vec{E}_2 = \frac{1}{4\pi\epsilon_0} \frac{-2}{(z^2 + 9)^{\frac{3}{2}}} (\hat{z}z + \hat{x}3)$$

add up ; superposition

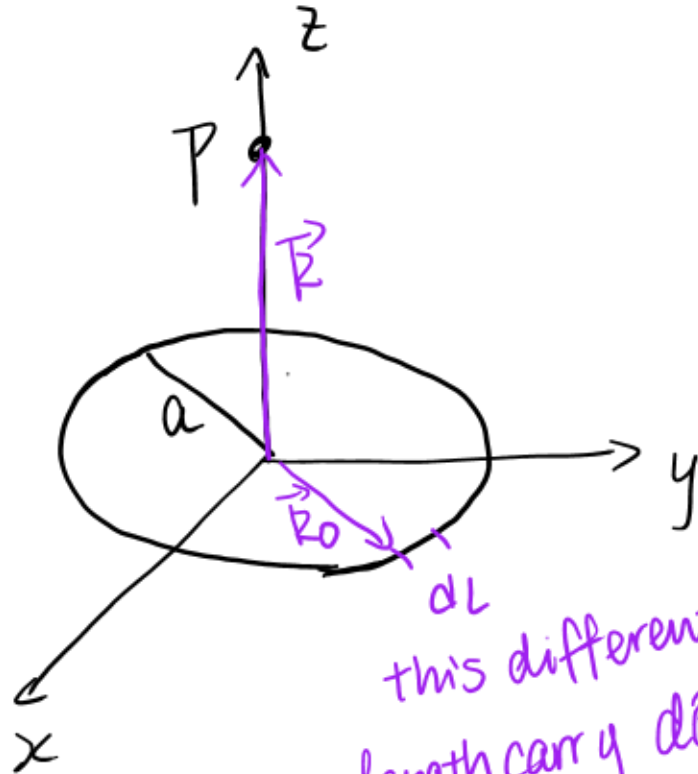
$$\vec{E}_{total} = \vec{E}_1 + \vec{E}_2 = \frac{2}{4\pi\epsilon_0} \frac{1}{(z^2 + 9)^{\frac{3}{2}}} (-\hat{x}6) = -\hat{x} \frac{3}{\pi\epsilon_0} \frac{1}{(z^2 + 9)^{\frac{3}{2}}}$$

Calculate the electric field at point P from line charge distribution [Example]

Use Coulomb's law and the principle of superposition to calculate the electric field $\vec{E}$ on the axis of a thin ring with radius a and uniform line charge density ρ_L .

☐ Choose cylindrical coordinate system.

$$\vec{E} = \frac{1}{4\pi\epsilon} \frac{Q}{|\vec{R} - \vec{R}_0|^3} (\vec{R} - \vec{R}_0)$$



this differential length carry $dQ = \rho_L \cdot dl$
 $dQ = \rho_L \cdot a \cdot d\phi$
 $dl = a d\phi$

For each differential length $^{(dl)}$ along the ring the $\vec{E}$ at P is

$$d\vec{E} = \frac{1}{4\pi\epsilon} \frac{dQ}{(z^2 + a^2)^{\frac{3}{2}}} (\hat{z}z - \hat{r}a)$$

$\left. \begin{array}{l} \vec{R} = \hat{z}z \\ \vec{R}_0 = \hat{r}a \\ \vec{R} - \vec{R}_0 = (\hat{z}z - \hat{r}a) \end{array} \right\}$

$$= \frac{1}{4\pi\epsilon} \frac{\rho_L \cdot a \cdot d\phi}{(z^2 + a^2)^{\frac{3}{2}}} (\hat{z}z - \hat{r}a)$$

(see next page)

Calculate the electric field at point P from line charge distribution [Example]

Use Coulomb's law and the principle of superposition to calculate the electric field $\vec{E}$ on the axis of a thin ring with radius a and uniform line charge density ρ_L .

integrate $d\vec{E}$ produced by all $\rho_L \cdot dL$ along the ring $\phi \in [0, 2\pi)$

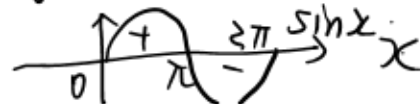
$$\vec{E} = \int_{\phi=0}^{\phi=2\pi} \frac{1}{4\pi\epsilon} \frac{\rho_L a d\phi}{(z^2 + a^2)^{\frac{3}{2}}} (\hat{z}z - \hat{r}a)$$

We would expect $\vec{E}_{\text{total}}$ along $\hat{z}$ only

$$\text{Therefore } \vec{E} = \frac{1}{4\pi\epsilon} \frac{\rho_L a \cdot 2\pi}{(z^2 + a^2)^{\frac{3}{2}}} \hat{z}z = \hat{z} \frac{\rho_L a z}{2\epsilon (z^2 + a^2)^{\frac{3}{2}}}$$

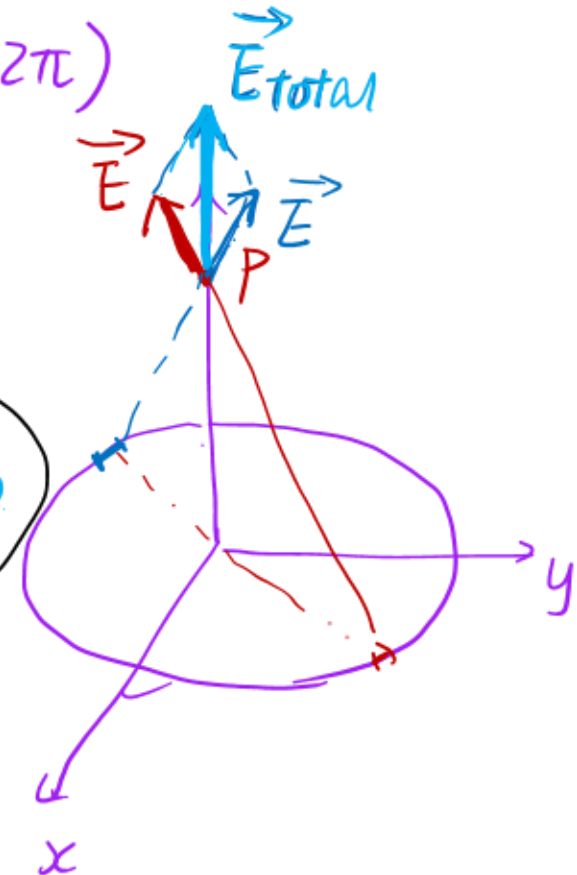
Prove: $\hat{r} = \hat{x}\cos\phi + \hat{y}\sin\phi$

$$\int_{\phi=0}^{\phi=2\pi} (\hat{x}\cos\phi + \hat{y}\sin\phi) d\phi = 0$$



integrating this part should be 0 in this problem

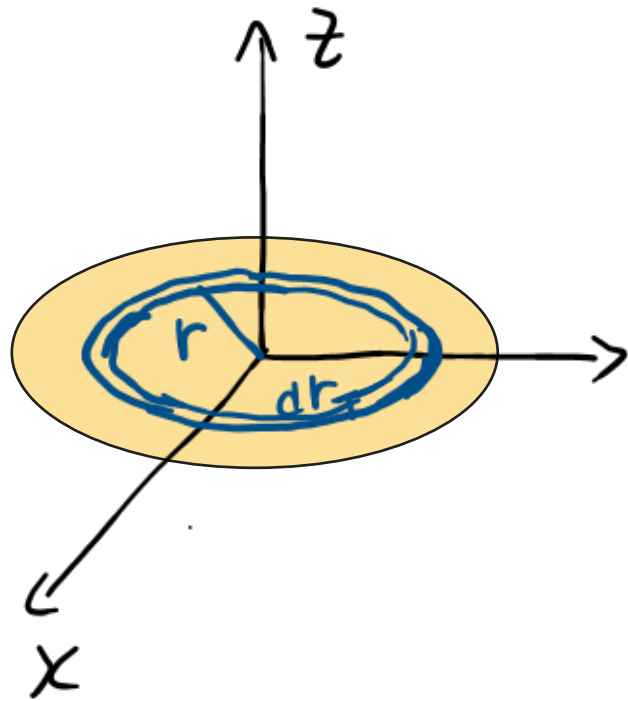
Final



Calculate the electric field at point P from surface charge distribution [Example]

Building on the expression obtained in the previous example, find the electric field at point P with Cartesian coordinates $(0, 0, z)$ from a circular disk of radius a and uniform surface charge density ρ_s in the x - y plane.

Then, evaluate the electric field from an infinite sheet with uniform surface charge density ρ_s by letting $a \rightarrow \infty$.



Previous
$$\vec{E} = \hat{z} \frac{z \rho_s a}{2\epsilon (z^2 + a^2)^{3/2}}$$

Here, decompose the disk into many concentric rings.

for each ring. $\rho_L = \rho_s dr$

$$d\vec{E} = \hat{z} \frac{z \rho_s dr \cdot r}{2\epsilon (z^2 + r^2)^{3/2}}$$

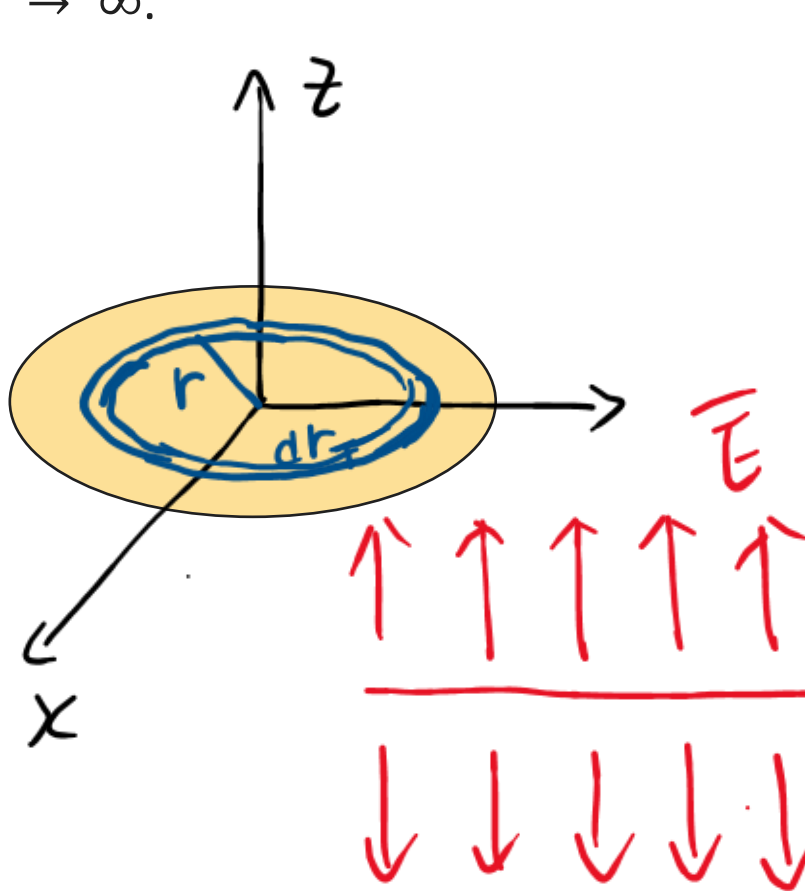
u-substitution

$$\vec{E} = \int_{r=0}^{r=a} \hat{z} \frac{z \rho_s dr \cdot r}{2\epsilon (z^2 + r^2)^{3/2}} = \hat{z} \frac{z \cdot \rho_s}{2\epsilon} \left(\int_0^a \frac{r dr}{(z^2 + r^2)^{3/2}} \right) = \hat{z} \frac{z \rho_s}{2\epsilon} \left(\frac{1}{\sqrt{z^2}} - \frac{1}{\sqrt{z^2 + a^2}} \right)$$

Calculate the electric field at point P from surface charge distribution [Example]

Building on the expression obtained in the previous example, find the electric field at point P with Cartesian coordinates $(0, 0, z)$ from a circular disk of radius a and uniform surface charge density ρ_s in the x - y plane.

Then, evaluate the electric field from an infinite sheet with uniform surface charge density ρ_s by letting $a \rightarrow \infty$.



$$\vec{E} = \hat{z} \frac{z \rho_s}{2\epsilon} \left(\frac{1}{\sqrt{z^2}} - \frac{1}{\sqrt{z^2 + a^2}} \right)$$

when $a \rightarrow \infty$

$$\vec{E} = \hat{z} \frac{z \rho_s}{2\epsilon} \frac{1}{|z|} = \begin{cases} \hat{z} \frac{\rho_s}{2\epsilon}, & z > 0 \\ -\hat{z} \frac{\rho_s}{2\epsilon}, & z < 0 \end{cases}$$

$\rho_s > 0$

uniform $\Rightarrow$

(in class notes)

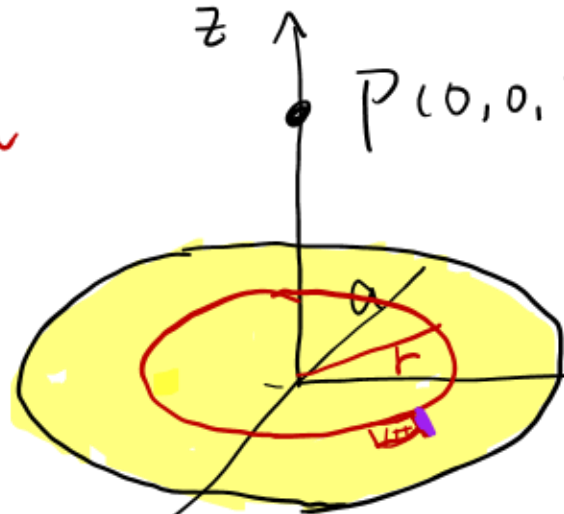
Calculate the electric field at point P from surface charge distribution [Example]

$$\int x^n dx = \frac{x^{n+1}}{n+1} \Big|_{x_1}^{x_2}$$



Building on the expression obtained in the previous example, find the electric field at point P with Cartesian coordinates $(0, 0, z)$ from a circular disk of radius a and uniform surface charge density ρ_s in the x - y plane.

Then, evaluate the electric field from an infinite sheet with uniform surface charge density ρ_s by letting $a \rightarrow \infty$.



$$\vec{E} = \hat{z} \frac{z \rho_s \pi a^2}{2 \epsilon (z^2 + a^2)^{3/2}}$$

$$u = z^2 + r^2$$

$$du = 2r dr$$

$$r dr = \frac{du}{2}$$

$$\rho_s > 0 \Rightarrow \rho_s = \rho_s \cdot dA$$

$$\int u = z^2 + a^2 \quad \frac{du}{2u^{3/2}}$$

$$= \int_{u=z^2}^{u=z^2+a^2} \frac{du}{2u^{3/2}} = \left[-\frac{1}{u^{1/2}} \right]_{z^2}^{z^2+a^2} = -\frac{1}{\sqrt{z^2+a^2}} + \frac{1}{\sqrt{z^2}}$$

$$d\vec{E} = \hat{z} \frac{z \cdot \rho_s dA \cdot r}{2 \epsilon (z^2 + r^2)^{3/2}}$$

$$\vec{E} = \int_{r=0}^{r=a} \hat{z} \frac{z \rho_s r dr}{2 \epsilon (z^2 + r^2)^{3/2}}$$

$$= \hat{z} \frac{z \rho_s}{2 \epsilon} \int_0^a \frac{r dr}{(z^2 + r^2)^{3/2}}$$

$$= -\frac{1}{\sqrt{z^2+a^2}} + \frac{1}{\sqrt{z^2}}$$

$$dQ = \rho_s \cdot dA = \rho_s \cdot r dr d\phi$$

$$= \rho_s \cdot dA = \rho_s \cdot r dr d\phi$$

$$\text{sgn}(z) = \begin{cases} +1 & z > 0 \\ -1 & z < 0 \end{cases}$$

$$\vec{E} = \hat{z} \frac{z \rho_s}{2 \epsilon} \left(\frac{1}{\sqrt{z^2}} - \frac{1}{\sqrt{z^2+a^2}} \right)$$

$$\frac{z}{|z|} = \begin{cases} +1 & z \geq 0 \\ -1 & z < 0 \end{cases}$$

$$\rho_s = \frac{dQ}{dA}$$

$$\rho_s = \frac{dQ}{dA}$$

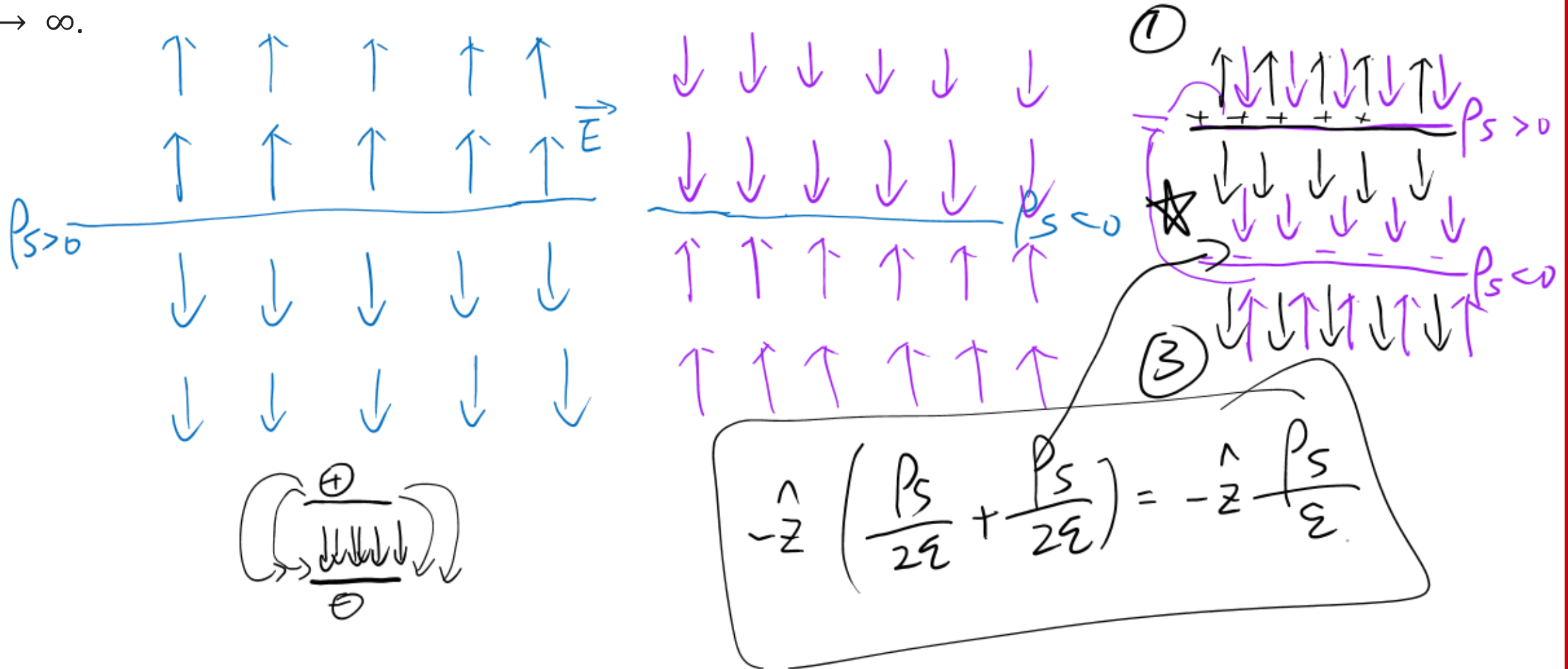
$$a \rightarrow \infty$$

$$\vec{E}_{\text{total}} = \hat{z} \frac{z \rho_s}{2 \epsilon} \frac{1}{|z|} = \hat{z} \frac{\rho_s}{2 \epsilon} \text{sgn}(z)$$

Calculate the electric field at point P from surface charge distribution [Example]

Building on the expression obtained in the previous example, find the electric field at point P with Cartesian coordinates $(0, 0, z)$ from a circular disk of radius a and uniform surface charge density ρ_s in the x - y plane.

Then, evaluate the electric field from an infinite sheet with uniform surface charge density ρ_s by letting $a \rightarrow \infty$.



conceptual geometry of an electrostatic actuator

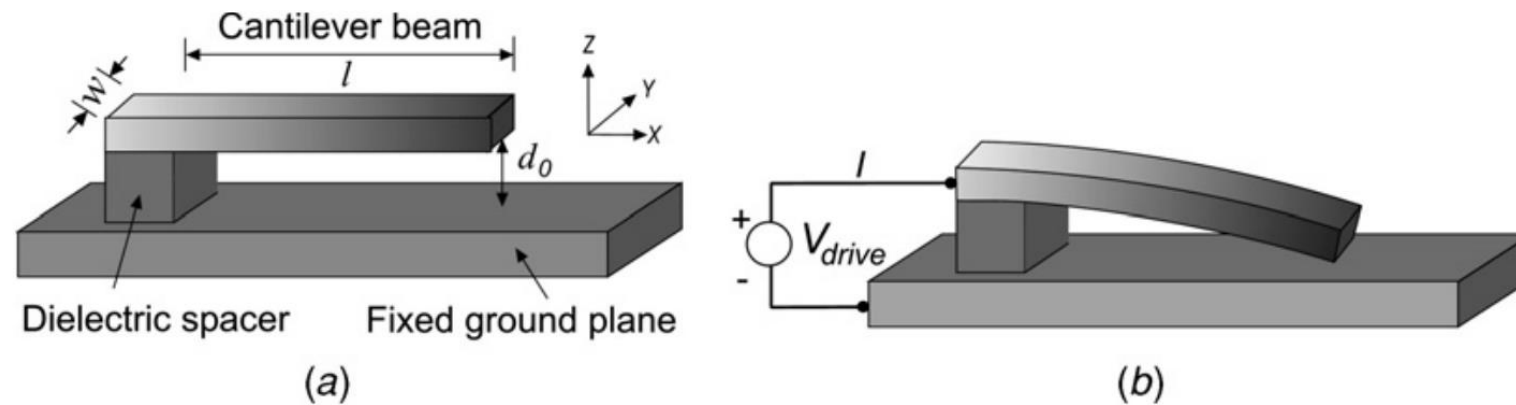


Figure 3. (a) A cantilever beam separated from a fixed ground plane by a dielectric spacer. (b) A conceptual diagram showing the deformation profile of the beam due to an electrostatic force when biased oppositely using a constant drive voltage.